Multivariate Analysis

Canonical Correlation Analysis

Johnson, RA & Wichern, DW. "Applied Multivariate Analysis", Chapter 10, 6th edition, Pearson International Edition, 2007.

Canonical Correlation analysis aims to describe relationships between two sets of variables, $\boldsymbol{X}^{(1)}_{p\times 1}$ and $\boldsymbol{X}^{(2)}_{q\times 1}$.

Examples include the association between indicators of arithmetic speed and indicators of arithmetic power; or university performance and pre-university achievement. It does so by focusing on the correlation between LINEAR combinations of variables in one set to LINEAR combinations of variables in another set. Hence in general the analysis involves

Determine the pair of linear combinations having the largest correlation,

$$U_1 = a_{11}X_1^{(1)} + a_{12}X_2^{(1)} + \cdots + a_{1p}X_p^{(1)}$$

and

$$V_1 = b_{11}X_1^{(2)} + b_{12}X_2^{(2)} + \cdots + b_{1q}X_q^{(2)}.$$

Determine the pair of linear combinations having the largest correlation among all pairs uncorrelated with the initial pair.

And so on.

The pairs of linear combinations are called CANONICAL VARIABLES and their correlations are called CANONICAL CORRELATIONS, which measure the strength of association between the two sets of variables.

The overall aim is to attempt to concentrate a high-dimensional relationship between two sets of variables into a few pairs of canonical variables.

Objective in terms of maximizing correlation matrix

Given the covariance matrix between the variables within a group and between variables from different groups,

$$\boldsymbol{\Sigma} = \begin{bmatrix} \boldsymbol{\Sigma}_{11(p\times p)} & \vdots & \boldsymbol{\Sigma}_{12(p\times q)} \\ \cdots & \vdots & \cdots \\ \boldsymbol{\Sigma}_{21(q\times p)} & \vdots & \Sigma_{22(q\times q)} \end{bmatrix}$$

Note the partitioning of the covariance matrix in terms of the 2 subsets of variables.

Let the linear combinations be $\boldsymbol{U} = \boldsymbol{a}'\boldsymbol{X}^{(\mathbf{1})}$ and $\boldsymbol{V} = \boldsymbol{b}'\boldsymbol{X}^{(\mathbf{2})}$ then

$$Var(\boldsymbol{U}) = \boldsymbol{a}'\boldsymbol{\Sigma}_{\mathbf{11}}\boldsymbol{a}$$
$$Var(\boldsymbol{V}) = \boldsymbol{b}'\boldsymbol{\Sigma}_{\mathbf{22}}\boldsymbol{b}$$
$$Covar(\boldsymbol{U},\boldsymbol{V}) = \boldsymbol{a}'\boldsymbol{\Sigma}_{\mathbf{12}}\boldsymbol{b}$$

These expressions just follow the normal rules for finding the variance of a random variable pre-multiplied by a constant, e.g. Var(aX)=a^2Var(X), etc.

We wish to find vectors $\boldsymbol{a}$ and $\boldsymbol{b}$ such that

$$Corr(\boldsymbol{U},\boldsymbol{V}) = \frac{\boldsymbol{a}'\boldsymbol{\Sigma}_{\mathbf{12}}\boldsymbol{b}}{\sqrt{\boldsymbol{a}'\boldsymbol{\Sigma}_{\mathbf{11}}\boldsymbol{a}}\sqrt{\boldsymbol{b}'\boldsymbol{\Sigma}_{\mathbf{22}}\boldsymbol{b}}}$$

And this is of course how we calculate the correlation as the ratio of the covariance divided by the product of the 2 sd's.

is as large as possible.

Solution to maximization problem

It can be shown that the $\boldsymbol{a}$ and **b** that will maximize the $Corr(U, V) = \rho_1^*$ are given by

$$U_1 = \underbrace{\boldsymbol{e}_1'\boldsymbol{\Sigma}_{11}^{-1/2}}_{\boldsymbol{a}_1'}\boldsymbol{X}^{(1)} \text{ and } V_1 = \underbrace{\boldsymbol{f}_1'\boldsymbol{\Sigma}_{22}^{-1/2}}_{\boldsymbol{b}_1'}\boldsymbol{X}^{(2)}$$

and for the k^{th} pair,

$$U_k = \underbrace{\boldsymbol{e}_k'\boldsymbol{\Sigma}_{11}^{-1/2}}_{\boldsymbol{a}_k'}\boldsymbol{X}^{(1)} \text{ and } V_k = \underbrace{\boldsymbol{f}_k'\boldsymbol{\Sigma}_{22}^{-1/2}}_{\boldsymbol{b}_k'}\boldsymbol{X}^{(2)}$$

So similar to PCA but scaled by $\Sigma_{11}^{-1/2}$ or $\Sigma_{22}^{-1/2}$

where $\rho_1^{*2} \geq \rho_2^{*2} \geq \cdots \geq \rho_p^{*2}$ are the eigenvalues of

$$\boldsymbol{R} = \boldsymbol{\Sigma}_{11}^{-1/2}\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21}\boldsymbol{\Sigma}_{11}^{-1/2}$$

We will unpack this product a little more later.

and $\boldsymbol{e}_1, \boldsymbol{e}_2, \dots, \boldsymbol{e}_p$ are the associated eigenvectors.

Equivalently, $\rho_1^{*2} \geq \rho_2^{*2} \geq \cdots \geq \rho_p^{*2}$ are the eigenvalues of

$$\boldsymbol{R} = \boldsymbol{\Sigma}_{22}^{-1/2}\boldsymbol{\Sigma}_{21}\boldsymbol{\Sigma}_{11}^{-1}\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1/2}$$

So you only need to calculate the eigendecomposition of one product and then you can obtain the f_i from the e_i. You will still need to divide by variance to ensure unit variance.

and associated eigenvalues $\boldsymbol{f}_1, \boldsymbol{f}_2, \dots, \boldsymbol{f}_p$ and each $\boldsymbol{f}_i$ is proportional to $\boldsymbol{\Sigma}_{22}^{-1/2}\boldsymbol{\Sigma}_{21}\boldsymbol{\Sigma}_{11}^{-1/2}\boldsymbol{e}_i$. Needs standardization so as to have unit variance – see r-code later on.

Properties of Canonical Variates

The canonical covariates have the following properties:

$$Var(U_k) = Var(V_k) = 1$$
$$Cov(U_k U_l) = Corr(U_k U_l) = 0$$
$$Cov(V_k V_l) = Corr(V_k V_l) = 0$$
$$Cov(U_k V_l) = Corr(U_k V_l) = 0$$

for $k \neq l$.

Standardized unit variances and independence among U and V of different linear combinations when k ne l.

If the original variables are standardized, the canonical variates are of the form

$$U_k = \boldsymbol{a}'_k \boldsymbol{Z}^{(1)} = \boldsymbol{e}'_k \boldsymbol{\rho}_{11}^{-1/2} \boldsymbol{Z}^{(1)} \text{ and } V_k = \boldsymbol{b}'_k \boldsymbol{Z}^{(2)} = \boldsymbol{f}'_k \boldsymbol{\rho}_{22}^{-1/2} \boldsymbol{Z}^{(2)}$$

Here Z represent the standardized variables and hence the covariance matrix of the Z is actually the correlation matrix of the X variables.

where

$\boldsymbol{\rho}_{11}$ and $\boldsymbol{\rho}_{22}$ are sub-matrices of the correlation matrix.

Interpretation

The canonical loadings can be thought of as the correlation between a variable set and its corresponding canonical variate and can thus be used to interpret the canonical variates, even though there is no information in these correlations about how one variable list contributes jointly to the canonical correlation with the other, i.e., they provide only univariate information.

The loadings can also be interpreted like regression coefficients, measuring the impact on the canonical variate of a one unit increase in the specific measured variable.

One way of thinking about Canonical Correlation Analysis is that every bit of correlation is wrung out of the original $X^{(1)}$ and $X^{(2)}$ variables and deposited in an orderly fashion into pairs of new variables U_k and V_k which have a special correlation structure.

Can also compute actual correlations between canonical variates and the original variables using $\boldsymbol{\rho}_{\boldsymbol{U},\boldsymbol{X}^{(1)}} = \boldsymbol{A}\boldsymbol{\Sigma}_{\mathbf{11}}\boldsymbol{V}_{\mathbf{11}}^{-\mathbf{1/2}}$ where $\boldsymbol{V}_{\mathbf{11}} = diag\left(Var\left(X_k^{(1)}\right)\right) = diag(\sigma_{kk})$ or $\boldsymbol{\rho}_{\boldsymbol{U},\boldsymbol{Z}^{(1)}} = \boldsymbol{A}_{\boldsymbol{z}}\boldsymbol{\rho}_{\mathbf{11}}$ where $\boldsymbol{A}_Z$ refers to canonical variates based on correlation matrix. Similarly,

$$\boldsymbol{\rho}_{\boldsymbol{V},\boldsymbol{X}^{(2)}} = \boldsymbol{B}\boldsymbol{\Sigma}_{\mathbf{22}}\boldsymbol{V}_{\mathbf{22}}^{-\mathbf{1/2}} \text{ or } \boldsymbol{\rho}_{\boldsymbol{V},\boldsymbol{Z}^{(2)}} = \boldsymbol{B}_{\boldsymbol{z}}\boldsymbol{\rho}_{\mathbf{22}}$$
$$\boldsymbol{\rho}_{\boldsymbol{U},\boldsymbol{X}^{(2)}} = \boldsymbol{A}\boldsymbol{\Sigma}_{\mathbf{12}}\boldsymbol{V}_{\mathbf{22}}^{-\mathbf{1/2}} \text{ or } \boldsymbol{\rho}_{\boldsymbol{U},\boldsymbol{Z}^{(2)}} = \boldsymbol{A}_{\boldsymbol{z}}\boldsymbol{\rho}_{\mathbf{12}}$$
$$\boldsymbol{\rho}_{\boldsymbol{V},\boldsymbol{X}^{(1)}} = \boldsymbol{B}\boldsymbol{\Sigma}_{\mathbf{21}}\boldsymbol{V}_{\mathbf{11}}^{-\mathbf{1/2}} \text{ or } \boldsymbol{\rho}_{\boldsymbol{V},\boldsymbol{Z}^{(1)}} = \boldsymbol{B}_{\boldsymbol{z}}\boldsymbol{\rho}_{\mathbf{21}}$$

where $\boldsymbol{V}$ and $\boldsymbol{\rho}$ are the covariance and correlation matrices of $\boldsymbol{X}$.

Interpretation of R

$$\boldsymbol{R} = \boldsymbol{\Sigma}_{11}^{-1/2}\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21}\boldsymbol{\Sigma}_{11}^{-1/2}$$

We are taking eigendecomposition of matrix of squared correlation coefficients as opposed to of covariance matrix which was the case for PCA.

When $p = 1$, i.e., just one $X^{(1)} = Y$,

R reduces to the squared correlation coefficient of $X^{(1)} = Y$ with the best linear predictor of $X^{(1)}$ using $X_1^{(2)}, X_2^{(2)}, \dots, X_q^{(2)}$

$$R = \rho^2_{Y.X_1^{(2)}\dots X_q^{(2)}} = \frac{\sigma_{YX}^T \Sigma_{XX}^{-1} \sigma_{XY}}{\sigma_Y^2}$$

The simplified form of R when we just have one variable in the first group.

And when both $p = 1$ and $q = 1$,

$R = \rho^2 = \frac{\sigma_{XY}^2}{\sigma_X^2 \sigma_Y^2}$, the squared correlation coefficient between X and Y.

Thus the j^{th} canonical correlation coefficient, ρ_j, can be interpreted as either the multiple correlation coefficient of V_j (the linear combination of $X^{(2)}$) with $X^{(1)}$ or of U_j (the linear combination of $X^{(1)}$) with $X^{(2)}$.

So ρ_j can be viewed as either

the proportion of the variance of U_j attributable to its linear regression on $X^{(2)}$

or of V_j attributable to its linear regression on $X^{(1)}$.

Example: CCA from first principles

```
#Read in correlation matrix
 > rho<-matrix(c(1,0.4,0.5,0.6,0.4,1.0,0.3,0.4,0.5,0.3,1.0,0.2,0.6,0.4,0.2,1.0),4,4,byrow=T)
 > rho
      [,1] [,2] [,3] [,4]
 [1,]  1.0  0.4  0.5  0.6
 [2,]  0.4  1.0  0.3  0.4
 [3,]  0.5  0.3  1.0  0.2
 [4,]  0.6  0.4  0.2  1.0


 #Partition the correlation matrix according to X1 and X2


 > rho11<-rho[1:2,1:2]
 > rho11
      [,1] [,2]
 [1,]  1.0  0.4
 [2,]  0.4  1.0

 > rho12<-rho[1:2,3:4]
 > rho12
      [,1] [,2]
 [1,]  0.5  0.6
 [2,]  0.3  0.4


 > rho21<-rho[3:4,1:2]
 > rho21
      [,1] [,2]
 [1,]  0.5  0.3
 [2,]  0.6  0.4

 > rho22<-rho[3:4,3:4]
 > rho22
      [,1] [,2]
 [1,]  1.0  0.2
 [2,]  0.2  1.0
```

Calculating matrices that form part of matrix product:

$$\rho_{11}^{-1/2}\rho_{12}\rho_{22}^{-1}\rho_{21}\rho_{11}^{-1/2}$$

and obtaining eigen-decomposition of this product.

```
> rho11inverse<-solve(rho11)
 > rho11inverse
           [,1]       [,2]
 [1,]  1.1904762 -0.4761905
 [2,] -0.4761905  1.1904762

 > Calculate square root of rho11inverse
 > a.eig <- eigen(rho11inverse)
 > rho11inverse.sqrt <- a.eig$vectors %*%
        diag(sqrt(a.eig$values)) %*% t(a.eig$vectors)

 > rho11inverse.sqrt
      [,1] [,2]
 [1,]  0.5  0.3
 [2,]  0.6  0.4
 >
 > rho22<-rho[3:4,3:4]
 > rho22
      [,1] [,2]
 [1,]  1.0  0.2
 [2,]  0.2  1.0
```

```
> rho22inverse<-solve(rho22)
 > a.eig <- eigen(rho22inverse)
 > rho22inverse.sqrt <- a.eig$vectors %*%
     diag(sqrt(a.eig$values)) %*%t(a.eig$vectors)

 > rho22inverse.sqrt
           [,1]       [,2]
 [1,]  1.0154525 -0.1025815
 [2,] -0.1025815  1.0154525

 > RHO<-
rho11inverse.sqrt%*%rho12%*%rho22inverse%*%rho21%*%
rho11inverse.sqrt

# Obtaining eigenvalue decomposition of RHO
 > RHO_eigen<-eigen(RHO)
 > RHO_eigen
 $values
 [1] 0.5457180317 0.0009089525
 $vectors
           [,1]       [,2]
 [1,] -0.8946536  0.4467605
 [2,] -0.4467605 -0.8946536
```

#Obtaining canonical loadings $,\boldsymbol{a_1} = \boldsymbol{e_k'\rho_{11}^{-1/2}}$

```
> a_1<-rho11inverse.sqrt%*%RHO_eigen$vectors[,1]
 > a_1<--1*a_1
 > a_1
 [1,] 0.8559647
 [2,] 0.2777371
```

Used fact that $\boldsymbol{b_1}$ proportional to $\boldsymbol{\rho_{22}^{-1}\rho_{21}a_1}$

```
 > b1_p<-rho22inverse%*%rho21%*%a_1
 > b1_p
 [1,] 0.4024674
 [2,] 0.5441802
```

$f_1 \propto \rho_{22}^{-1/2}\rho_{21}\rho_{11}^{-1/2}e_1$
$b_1 = \rho_{22}^{-1/2}f_1$
$\boldsymbol{b_1}$ proportional
to $\boldsymbol{\rho_{22}^{-1}\rho_{21}a_1}$

#Calculate covariance of V and form diagonal matrix of variances

```
 > V1_var<-1/(sqrt(t(b1_p)%*%rho22%*%b1_p))
 > V1_var<-matrix(c(V1_var,V1_var),2,1)
 > V1_var
 [1,] 1.353679
 [2,] 1.353679
```

$Var(V_1) = Var\left(b_1'Z^{(2)}\right) = b_1'\rho_{22}b_1$
To get this to be 1, you have to scale by this

Standardize b to unit variance

```
 > b_1<-b1_p * (V1_var) %Scaling b1 so that Var(V1)=Var(b1'Z^(2))=1
 > b_1
 [1,] 0.5448119
 [2,] 0.7366455
```

Calculate canonical correlation

```
 > rho_1<-sqrt(RHO_eigen$values[1])
 > rho_1
 [1] 0.7387273
```

```
#Tidy up

#Put canonical coefficients in matrices

> A_z<-t(matrix(a_1))
 > A_z
          [,1]      [,2]
 [1,] 0.8559647 0.2777371
 > B_z<-t(matrix(b_1))
 > B_z
          [,1]      [,2]
 [1,] 0.5448119 0.7366455

#Calculate correlations between canonical variates and measured variables

 > corr_U1Z1<-A_z%*%rho11
 > corr_U1Z1
          [,1]     [,2]
 [1,] 0.9670595 0.620123
 >
 > corr_V1Z2<-B_z%*%rho22
 > corr_V1Z2
         [,1]      [,2]
 [1,] 0.692141 0.8456079
 >

 > propvarexplained<-RHO_eigen$values[1]
 > propvarexplained
 [1] 0.545718
```

Sample Canonical Variates and Correlations

When we have a sample of n observations on each of the $p+q$ variables in $X^{(1)}$ and $X^{(2)}$ we work with the sample covariance matrix

$$\boldsymbol{S} = \begin{bmatrix} \boldsymbol{S}_{11(p\times p)} & \vdots & \boldsymbol{S}_{12(p\times q)} \\ \cdots & \vdots & \cdots \\ \boldsymbol{S}_{21(q\times p)} & \vdots & \boldsymbol{S}_{22(q\times q)} \end{bmatrix}$$

or the corresponding correlation matrix R.

Interpreting a Canonical Correlation Analysis

The canonical variates are artificial, they have no physical meaning.

The units of the canonical coefficients are proportional to the units of the variables in the two variable sets.

If the original variables are standardized (or, equivalently, we use the correlation matrices), the canonical coefficients have no units of measurement.

The canonical variates can be "identified" by looking at the correlations between the canonical variates and the original variables, (similar to a factor analysis).

Example 10.4 from Johnson & Wichern

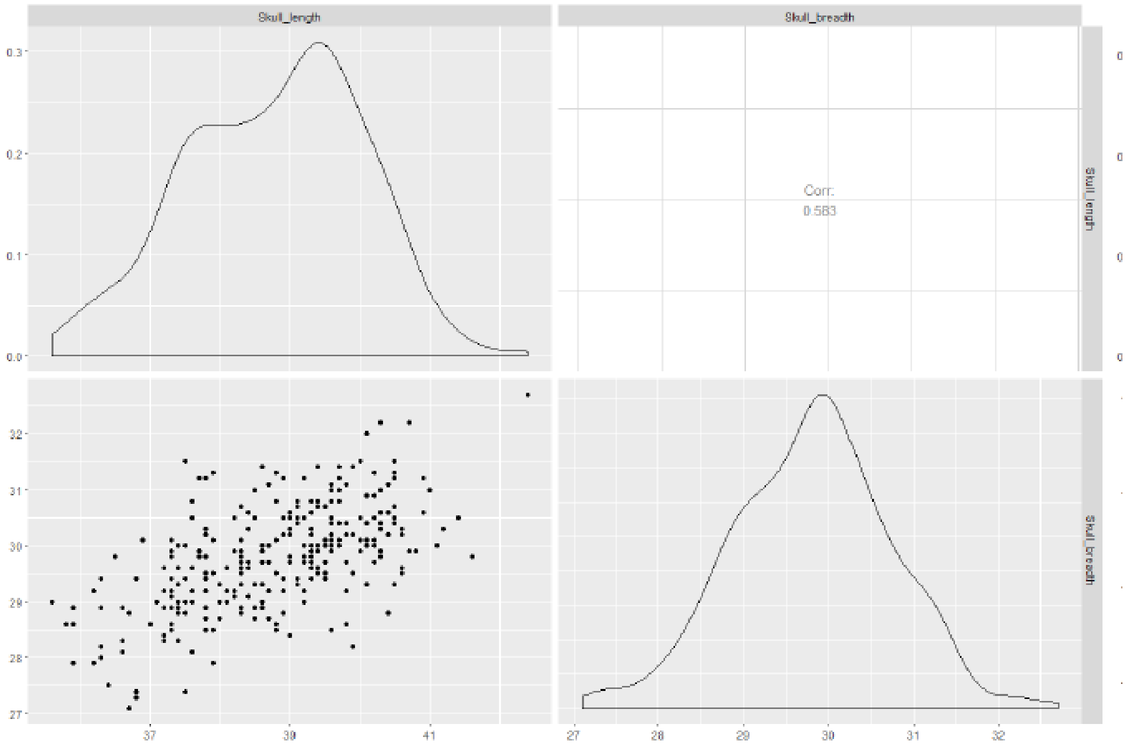
This example refers to measurements of bone and skull length and breadth of white leghorn fowl. Of interest is to find linear combinations of head and leg variables that are most strongly correlated.

```
> require(ggplot2)
> require(GGally)
> require(CCA)
> chickenbone<-read.table(file.choose())
> head(chickenbone)
   V1   V2   V3   V4    V5   V6   V7
1   3 40.3 31.0 80.3 116.5 78.6 73.8
2 219 41.0 31.0 77.4 119.0 74.7 70.8
3 147 40.3 30.3 84.5 125.5 79.3 73.6
4 152 38.4 29.6 74.0 109.0 71.9 66.4
5  82 38.3 29.7 76.8 113.0 74.0 69.0
6  96 39.4 30.0 82.6 123.0 74.4 70.5
> chickenbone<-chickenbone[,2:5]
> dimnames(chickenbone) <- list (1:nrow(chickenbone),
c("Skull_length","Skull_breadth","Femur_length","Tibia_length"))
>
> head(chickenbone)
  Skull_length Skull_breadth Femur_length Tibia_length
1         40.3          31.0         80.3        116.5
2         41.0          31.0         77.4        119.0
3         40.3          30.3         84.5        125.5
4         38.4          29.6         74.0        109.0
5         38.3          29.7         76.8        113.0
6         39.4          30.0         82.6        123.0
```

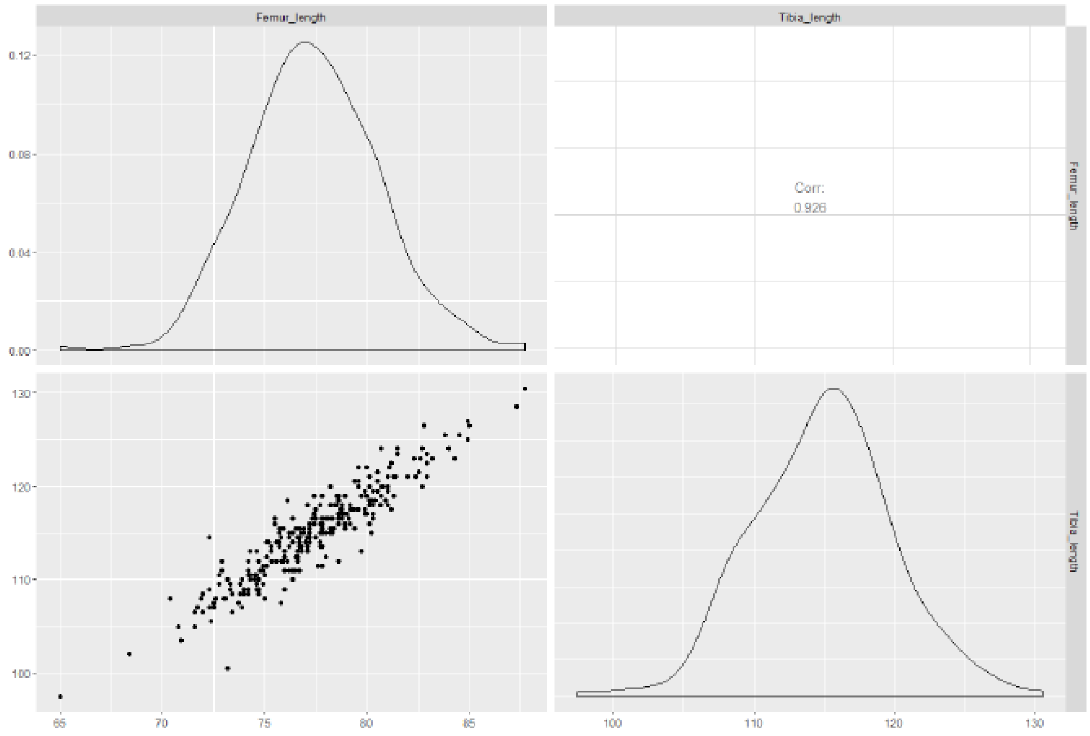
Variables we are interested in are V2 to V5

```
>skull <- chickenbone[, 1:2]
> bone <- chickenbone[, 3:4]
```

- ggpairs(skull)



> ggpairs(bone)



→ Relatively strong correlation among skull measurements

→ Very strong correlation among bone measurements as shown by narrow scatter of points along diagonal line

Examine Correlations

```
> matcor(skull, bone)
$Xcor
              Skull_length Skull_breadth
Skull_length     1.0000000     0.5830095
Skull_breadth    0.5830095     1.0000000

$Ycor
             Femur_length Tibia_length
Femur_length    1.0000000    0.9261046
Tibia_length    0.9261046    1.0000000

$XYcor
              Skull_length Skull_breadth Femur_length Tibia_length
Skull_length     1.0000000     0.5830095    0.5691105    0.6022595
Skull_breadth    0.5830095     1.0000000    0.5153099    0.5475994
Femur_length     0.5691105     0.5153099    1.0000000    0.9261046
Tibia_length     0.6022595     0.5475994    0.9261046    1.0000000
```

You can see the difference in strength of correlation between skull measurements and between bone measurements.

Correlations between skull and bone variables vary between 0.515 and 0.602.

```
> cc1 <- cc(skull, bone)

> cc1$cor
[1] 0.649681788 0.006499542

> # raw canonical coefficients
> cc1$xcoef
                    [,1]       [,2]
Skull_length  -0.5277691 -0.8288551
Skull_breadth -0.4925900  1.2284044
> cc1$ycoef
                    [,1]       [,2]
Femur_length -0.03788723 -0.8244348
Tibia_length -0.17753017  0.5004056

> # standardized skull canonical coefficients using diagonal matrix of skull sd's
> s1 <- diag(sqrt(diag(cov(skull))))
> s1 %*% cc1$xcoef
           [,1]      [,2]
[1,] -0.6610796 -1.038218
[2,] -0.4581002  1.142395

> # standardized bone canonical coefficients using diagonal matrix of bone sd's
> s2 <- diag(sqrt(diag(cov(bone))))
> s2 %*% cc1$ycoef
           [,1]      [,2]
[1,] -0.1216830 -2.647849
[2,] -0.8862546  2.498092
```

cc function in CCA library will perform a canonical correlation analysis

Relatively strong correlation between first pair of variates. Just about zero correlation between 2nd pair.

First canonical variate an average of two skull measurements. Second canonical variate seems to contrast length to breadth.

Tibia length has more influence on 1st CCV than femur length

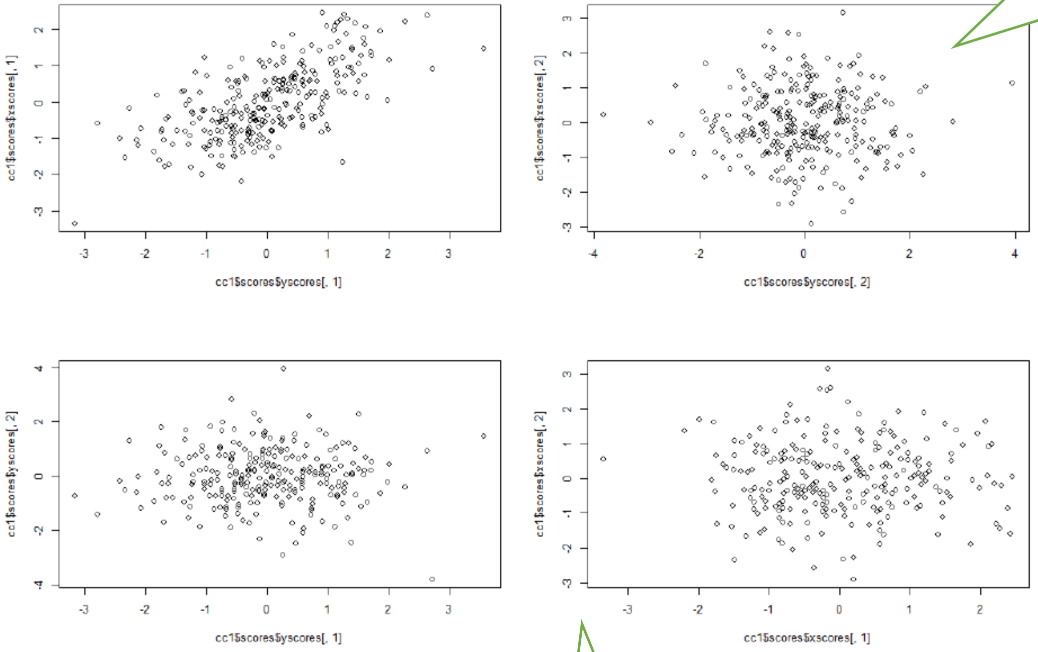
Impact of measured variables on canonical variates in sd units as opposed to measured units.

```
> # compute canonical loadings

> cc1$scores$xscores
              [,1]        [,2]
1   -1.393928961  0.208912317
2   -1.763367360 -0.371286283
3   -1.049115978 -0.650970778

273 -0.767641823  0.792086577
274 -0.862633739 -1.759510417
275  1.128817336  0.567235303
276 -0.067462108 -0.310501745
> cc1$scores$yscores
              [,1]          [,2]
1   -0.386149975 -1.6519301708
2   -0.720102441  1.9899447572
3   -2.143047912 -0.6109056238

272  0.590805793 -0.0289120137
273  1.010274433  0.3718158635
274 -0.584248221  0.5826608728
275 -0.113901724 -0.0912488436
276 -1.336044871  1.6774051147

> par(mfrow=c(2,2))
> plot(cc1$scores$yscores[,1], cc1$scores$xscores[,1])
> plot(cc1$scores$yscores[,2], cc1$scores$xscores[,2])
> plot(cc1$scores$yscores[,1], cc1$scores$yscores[,2])
> plot(cc1$scores$xscores[,1], cc1$scores$xscores[,2])
```

These are the actual values of the new canonical variates for each observation.

Shows the relatively strong correlation between the first pair of canonical variates

The very weak correlation between second pair of CCVs



And as expected, the zero correlation between the CCVs from different pairs

```
> # display canonical loadings
> cc1$scores$corr.X.xscores
                   [,1]       [,2]
Skull_length  -0.9281563 -0.3721905
Skull_breadth -0.8435159  0.5371043
```

Correlations between Skull variables and their CCVs

```
> cc1$scores$corr.X.yscores
                   [,1]         [,2]
Skull_length  -0.6030063 -0.002419068
Skull_breadth -0.5480169  0.003490932
```

Correlations between Skull variables and the CCVs based on bone variables

```
> cc1$scores$corr.Y.xscores
                  [,1]          [,2]
Femur_length -0.6122909 -0.0021731508
Tibia_length -0.6489968  0.0002983741
```

Correlations between bone variables and the CCVs based on skull variables

```
> cc1$scores$corr.Y.yscores
                  [,1]        [,2]
Femur_length -0.9424474 -0.33435446
Tibia_length -0.9989457  0.04590695
```

Correlations between bone variables and their CCVs

```
#From first principles:
> rho<-cor(chickenbone)
> rho
              Skull_length Skull_breadth Femur_length Tibia_length
Skull_length     1.0000000     0.5830095    0.5691105    0.6022595
Skull_breadth    0.5830095     1.0000000    0.5153099    0.5475994
Femur_length     0.5691105     0.5153099    1.0000000    0.9261046
Tibia_length     0.6022595     0.5475994    0.9261046    1.0000000
> rho11<-rho[1:2,1:2]
> rho11
              Skull_length Skull_breadth
Skull_length     1.0000000     0.5830095
Skull_breadth    0.5830095     1.0000000
>
> rho12<-rho[1:2,3:4]
> rho12
              Femur_length Tibia_length
Skull_length     0.5691105    0.6022595
Skull_breadth    0.5153099    0.5475994
> > rho21<-rho[3:4,1:2]
> rho21
             Skull_length Skull_breadth
Femur_length    0.5691105     0.5153099
Tibia_length    0.6022595     0.5475994

> rho22<-rho[3:4,3:4]
> rho22
             Femur_length Tibia_length
Femur_length    1.0000000    0.9261046
Tibia_length    0.9261046    1.0000000
```

Calculate correlation matrix and partition.

```
> rho11inverse<-solve(rho11)
> rho11inverse
             Skull_length Skull_breadth
Skull_length     1.5149221    -0.8832139
Skull_breadth   -0.8832139     1.5149221
>
> a.eig <- eigen(rho11inverse)
> rho11inverse.sqrt <- a.eig$vectors %*% diag(sqrt(a.eig$values)) %*% solve(a.eig$vectors
> rho11inverse.sqrt
           [,1]       [,2]
[1,]  1.1716962 -0.3768954
[2,] -0.3768954  1.1716962
>
> rho22inverse<-solve(rho22)
> rho22inverse
             Femur_length Tibia_length
Femur_length     7.025912    -6.506730
Tibia_length    -6.506730     7.025912
>
> a.eig <- eigen(rho22inverse)
> rho22inverse.sqrt <- a.eig$vectors %*% diag(sqrt(a.eig$values)) %*%solve(a.eig$vectors)
> rho22inverse.sqrt
[,1]      [,2]
[1,]  2.199609 -1.479065
[2,] -1.479065  2.199609
> RHO<-rho11inverse.sqrt%*%rho12%*%rho22inverse%*%rho21%*%rho11inverse.sqrt
> RHO
          [,1]      [,2]
[1,] 0.2500123 0.2073967
[2,] 0.2073967 0.1721164
```

```
> RHO_eigen<-eigen(RHO)
> RHO_eigen
eigen() decomposition
$values
[1] 4.220864e-01 4.224404e-05
$vectors
           [,1]       [,2]
[1,] -0.7695999  0.6385264
[2,] -0.6385264 -0.7695999
> a_1<-rho11inverse.sqrt%*%RHO_eigen$vectors[,1:2]
> a_1<--1*a_1
> a_1
          [,1]      [,2]
[1,] 0.6610796 -1.038218
[2,] 0.4581002  1.142395
> b1_p<-rho22inverse%*%rho21%*%a_1
> b1_p
                   [,1]        [,2]
Femur_length 0.07905521 -0.01720981
Tibia_length 0.57578345  0.01623646
> brhob<-t(b1_p)%*%rho22%*%b1_p
> brhob<-sqrt(diag(brhob))
> b11<-b1_p[,1]/brhob[1]
> b11
Femur_length Tibia_length
0.1216830    0.8862546
> b12<-b1_p[,2]/brhob[2]
> b12
Femur_length Tibia_length
-2.647849     2.498092
```

```
From CC analysis:
> # standardized psych canonical coefficients using
diagonal matrix of psych sd's
> s1 <- diag(sqrt(diag(cov(skull))))
> s1 %*% cc1$xcoef
           [,1]      [,2]
[1,] -0.6610796 -1.038218
[2,] -0.4581002  1.142395

> # standardized acad canonical coefficients using
diagonal matrix of acad sd's
> s2 <- diag(sqrt(diag(cov(bone))))
> s2 %*% cc1$ycoef
           [,1]      [,2]
[1,] -0.1216830 -2.647849
[2,] -0.8862546  2.498092
```

CCV coefficients obtained from first principles same as standardized coefficients from cc function.

Class Exercise 1

Use R and program from first principles to calculate the first pair of canonical variates of the covariance matrix and the canonical correlation.

```
> Cov
     [,1] [,2] [,3] [,4]
[1,]  100 0.00 0.00    0
[2,]    0 1.00 0.95    0
[3,]    0 0.95 1.00    0
[4,]    0 0.00 0.00  100
```

Save annotated r-code and output in a pdf file labelled studentno_CCA1.pdf and submit via VULA.

Class Exercise 2

Return to the Stock Price data and perform a canonical correlation analysis with $X^{(1)}$ being the banking rates of returns and $X^{(2)}$ being the rates of returns for the oil companies.

Save annotated r-code and output in a pdf file labelled studentno_CCA2.pdf and submit via VULA.

To Summarize (What you should have learnt):

- Definition of Canonical Correlation Analysis (CCA), Canonical variates (CCVs) and Canonical Correlation (Slide1)
- Partitioning of the covariance matrix and calculation of correlation between 2 linear combinations (Slide 2)
- Expression for canonical coefficients and introduction of R and relationship between e and f (Slide 3)
- Properties of U and V and formulation in terms of correlation matrix (standardized variables) (Slide 4)
- Guide to interpretation and distinction between canonical coefficients and correlations between measured variables and canonical variables (Slide 5)
- More about R (Slide 6)
- R-code for CCA from first principles (Slides 7 to 10)
- Example using R function cc from R library CCA (Slides 13 to 18)